

PECTORAL GIRDLE DEVELOPMENT

The pectoral girdle develops at an interface between the tissues forming the head and neck, the back and the upper limb. Each of these regions appears to have different cell lineages and genes contributing to the development of similar tissues. Studies on the development of the avian scapula, also confirmed in the mouse, show that lateral scapular structures derive from the somatopleuric mesenchyme, whereas the medial border of the scapula and the attachment sites for the rhomboid muscles originate from dermomyotomes of somites: specifically, the hypaxial domain (Wang et al 2005, Valasek et al 2010, Valasek et al 2011). The connective tissues of those muscles and their attachment sites on the pectoral girdle innervated by the accessory nerve are derived from neural crest (Matsuoka et al 2005).

Early limb development is patterned by somatopleuric mesenchyme, within a permissive interaction with the limb ectoderm. Upper limb myoblasts migrate laterally from the ventrolateral border of cervical dermomyotomes and form dorsal and ventral masses prior to migration into the limb bud. After a brief period within the limb bud, a subset of the myoblasts migrate medially out of the limb bud and into the axial mesenchyme to form the pectoral muscles ventrally and latissimus dorsi dorsally. This origin of superficial pectoral girdle muscles has been described as an 'in-out' mechanism (Valasek et al 2011), and refers to the initial migration into the limb bud and the subsequent migration back towards axial structures. Only myoblasts undergo this second, medial, migration out of the limb bud; the somatopleuric mesenchyme remains within the proximal limb anlagen and produces the gleno-humeral joint and coracoid process.

Some myoblasts, originating from the ventrolateral edges of cervical dermomyotomes and extending as the hypaxial domain of the somite, switch off their myogenic fate and follow a cartilage lineage to form the medial part of the scapula. The remaining hypaxial domain myoblasts form serratus anterior, and the rhomboid muscles, deep pectoral girdle muscles, attach to this medial part (Valasek et al 2011).

Trapezius and sternocleidomastoid, innervated by the accessory nerve, are derived from the upper occipital somites (see Fig. 36.7); they migrate into neural crest mesenchyme, which forms their connective tissue. The proximal and distal attachments of trapezius, to the nuchal line of the occipital bone and the upper part of the spine of the scapula, respectively, are derived from post-otic neural crest cells (Matsuoka et al 2005). The attachment of sternocleidomastoid to the clavicle also demonstrates its dual origin from neural crest cells and somatopleuric mesenchyme. The sternum is derived from somatopleuric mesenchyme, although there is evidence of neural crest within the manubrium.

The 'muscle scaffold model' proposed by Matsuoka et al (2005), where muscle connective tissue and muscle attachment sites illustrate the origin of the mesenchymal cells patterning their development, may explain a range of anomalies based on defects of post-otic neural crest and which affect the pharynx and larynx as well as the shoulder region, e.g. Klippel-Feil disease, Sprengel's deformity and cleidocranial dysplasia.

UPPER LIMB DEVELOPMENT

The upper limb bud is first visible as a thickening along the lateral longitudinal axis of the body wall opposite somites 8–10, at the level of the entrance to the cranial intestinal portal during stage 12. It enlarges, protruding laterally from its elliptical base at the body wall as a symmetrical bud, with a curved border and an apical ectodermal ridge forming its distal tip. Initially, it has equal and relatively flat dorsal and ventral ectodermal surfaces and a somatopleuric mesenchymal core. By stage 13, the upper limb bud is curved ventrally. By stage 14, it has

elongated and is opposite the developing ventricles of the heart; it is directed caudally because of preferential elongation of the preaxial part of the bud. At stage 15, the upper limb can be subdivided into definite regions. The proximal portion, still showing the dorsal bulge and ventral curve, is the shoulder and upper arm region; the next distal portion can now be identified as the forearm; the most distal portion is now expanded into a flattened hand plate.

At stage 16, there are the first indications of digit rays in the hand plate. The limb may be close to the body wall or may be abducted. By stage 17, the upper limb has an elbow region and well-defined digit rays; the hand plate may show a crenated rim, indicating the start of tissue removal between the digits (see Fig. 15.1). Crenation of the hand plate between the digit rays continues in stage 18 embryos (44 days). Changes during stages 19–23 are concerned with growth of the limb and separation of the digits. The hands now curve over the cardiac region. The distal phalangeal portions of the fingers enlarge at stage 21, forming the nail beds. For more details, see O'Rahilly and Gardner (1975), Gupta et al (2000).

Most of the bones in the appendicular skeleton are derived from somatopleuric mesenchyme. Within the upper limb, although the clavicle and coracoid portion of the scapula arise from somatopleuric mesenchyme, studies in both avian and mouse embryos have shown that the distal third of the blade of the scapula is derived from the somites (Shearman et al 2011, Valasek et al 2010). Prechondroblasts are present in the upper limb at stage 13, and condensations of cartilage can be detected at stage 16, when the humeral anlage can be recognized. Cavitation of the shoulder and elbow joints occurs at 7–8 weeks. By stage 17, when the radius and ulna chondrify, the branched tips of the radial, median and ulnar nerves have migrated to the distal hand plate. The carpals chondrify at stage 18, when the hand plate shows notching of the digital rays.

VESSELS

Arteries

In the upper limb, usually only one arterial trunk, the subclavian, persists; it probably represents the lateral branch of the seventh intersegmental artery. Its main continuation, the axillary artery of the upper limb, becomes the axillary and brachial arteries, passes into the forearm deep to the flexor muscle mass and terminates as a deep plexus in the developing hand (Fig. 47.1). The original axial vessel ultimately persists as the anterior interosseous artery and the deep palmar arch. A branch from the main trunk passes dorsally between the early radius and ulna as the posterior interosseous artery; a second branch accompanies the median nerve into the hand, where it ends in a superficial capillary plexus. The radial and ulnar arteries are the latest arteries to appear in the forearm. Initially, the radial artery arises more proximally than the ulnar artery, crosses anterior to the median nerve, and supplies biceps brachii. Later, the radial artery establishes a new connection with the main trunk at or near the level of origin of the ulnar artery, and the upper portion of its original stem usually disappears. On reaching the hand, the ulnar artery links up with the superficial palmar plexus, from which the superficial palmar arch is derived. The median artery commonly loses its distal connections and is reduced to a small vessel. The radial artery passes to the dorsal surface of the hand, gives off dorsal digital branches, traverses the first intermetacarpal space and joins the deep palmar arch.

Anomalies of the arterial tree in the forelimb are fairly common: there are several potential causes. The arteries have multiple and plexiform sources; the principal arteries, anastomoses and periarticular networks emerge according to a temporal sequence; and some paths that are initially functionally dominant subsequently regress. In general,

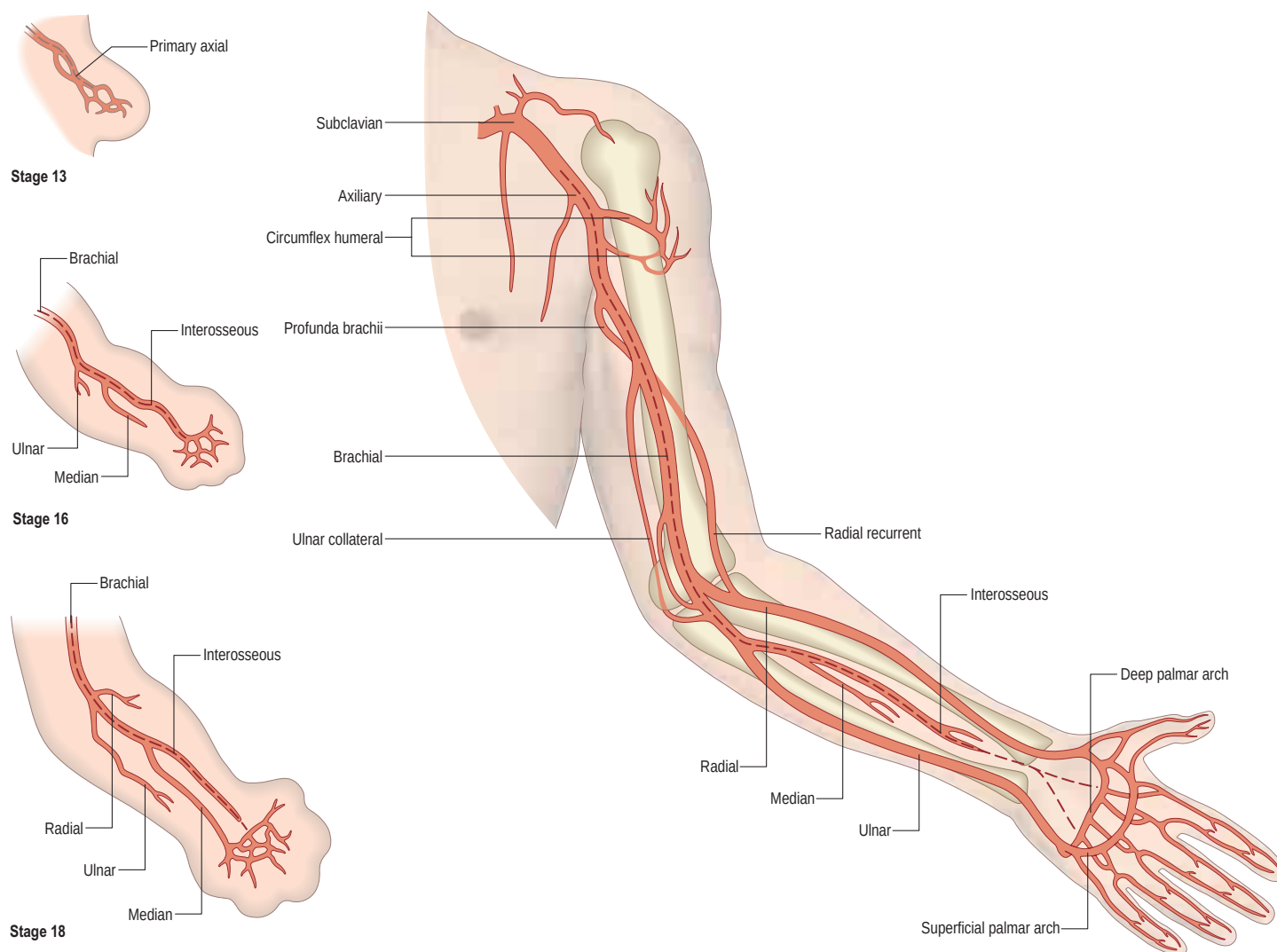


Fig. 47.1 Stages in the development of the arteries of the arm. The path of the axial artery is indicated by an interrupted line.

anomalous patterns may present as differences in the mode and proximal-distal level of branching; the presence of unusual compound arterial segments; aberrant vessels that connect with other principal vessels, arcades or plexuses; vessels that occupy exceptional tissue planes (e.g. superficial fascia instead of the usual subfascial route), or vessels that have unexpected neural, myological or osteoligamentous relationships.

Veins

In the upper limb, the preaxial vein becomes the cephalic vein and drains into the axillary vein at the shoulder. The postaxial vein becomes the basilic vein, which passes deep in the arm to continue as the axillary vein.

ULTRASOUND ANTENATAL IMAGING

In the routine 18–20-week ultrasound fetal assessment, shortening of long bones below 2 standard deviations should lead to a full assessment of all bones (McHugo 2007). Whereas the majority of lethal short limb dysplasias show marked shortening by this time, non-lethal short limb dysplasias seen at birth are not apparent at 20 weeks' gestation. Normally, the humerus should be slightly longer than the radius and ulna. The term rhizomelia is used where limb shortening affects the proximal bones more than the distal bones.

NEONATAL UPPER LIMB

In general, the upper limbs in the neonate are proportionately shorter than they are in the adult. They are long compared with the neonatal trunk and lower limbs, and extend to the upper thigh as they do in the adult, but the trunk is much shorter in the neonate (see Fig. 14.8). At birth, the upper limbs are about the same length as the lower limbs,

but much more developed. When the proportions of parts of the upper limb are examined, the forearm is longer than the arm in the newborn, and more so in boys than girls. Only primary centres of ossification are present in the upper limb, apart from a centre in the head of the humerus. The elbow of the newborn can flex to 145° but lacks some 10–15% of full extension. The neonate has a relatively strong grasp within the first few days. The fingernails usually extend to the fingertips or just beyond. They are soft at birth but soon dry and become quite firm and sharp.

Peripheral arterial puncture in the neonate

It is common practice to insert a small-bore cannula into a peripheral artery in neonates receiving intensive care when either the umbilical artery is not accessible or there are clinical reasons to avoid cannulation of the umbilical vessels. Trans-illumination can be used to provide an outline of the artery to be cannulated. The peripheral arteries that are most commonly used are the radial artery, just above the anterior surface of the wrist, and the posterior tibial artery, posterior to the medial malleolus. The proximity of the ulnar nerve to the ulnar artery increases the risk of nerve damage associated with arterial cannulation of the ulnar artery, and the relatively poor collateral circulation associated with the dorsalis pedis artery means that this artery is used only as a last resort. The brachial artery at the antecubital fossa also has a poor collateral circulation and the median nerve is in close proximity; it is generally considered, therefore, that cannulation of this artery is not justified.

Confirmation that an adequate collateral circulation is present when cannulating the radial artery can be obtained by performing Allen's test, in which both the radial and ulnar arteries are compressed at the wrist after exsanguination of the hand; release of pressure on the ulnar artery while maintaining occlusion of the radial artery should result in reperfusion of the hand if an adequate collateral ulnar arterial supply is present. Alternatively, intact arterial flow can usually be confirmed, particularly in the preterm infant, by direct visualization of the arteries

using trans-illumination. A cold light source is placed on the posterior aspect of the lower forearm and the shadow of the pulsating arteries can be seen on the anterior surface of the forearm.

DEVELOPMENTAL ANOMALIES OF THE UPPER LIMB

Approximately 1 in 600 neonates will have a congenital anomaly of the hand or forearm, 30% of which will be bilateral. They occur with an approximately equal frequency in males and females. A defect is twice as likely to occur in the upper limb as in the lower limb. The most critical period for the development of anomalies is during stages 12–16.

One classification of congenital limb malformation uses seven subgroups (Swanson 1976). These describe a clinical picture and do not always relate accurately to the developmental process occurring in the limb. The first subgroup is failure of formation of parts of a limb. The second subgroup, failure of differentiation, includes unsuccessful separation of parts, and so includes the range of syndactylies. Groups three to five include limb duplication, overgrowth and undergrowth. Group six includes limb amputations, mainly by adherent amniotic bands, and the last group includes all other generalized skeletal anomalies. A more recent classification of upper limb anomalies, based on developmental molecular biology and pathogenetics, clearly defines all terms used to describe anomalous structure and provides a comprehensive and consistent terminology for the range of anomalies (Oberg et al 2010, Tonkin et al 2013, Goldfarb 2015).

LIMB REDUCTION DEFECTS

Limb reduction defects are categorized as failures of development. The basis of limb reductions seen in infants whose mothers took thalidomide during their pregnancy seems likely to be due to death of mesenchyme cells in the early limb bud, causing phocomelia. Proximal segments of the limb were lost, although, in some cases, normal distal

segments still formed. There is evidence from experiments in chick embryos that thalidomide interferes with vascularization of the limbs (Vargesson 2009), and this could lead to mesenchymal cell death. Recent work has identified a protein to which thalidomide binds and could eventually lead to uncovering molecular mechanisms of thalidomide teratogenicity (Ito et al 2010).

Inheritance of autosomal dominant genes may cause anomalies of the hand. This could, for example, result in absence of the central digit ray of the distal hand plate and subsequent absence of the middle or ring finger and its metacarpals. Extra preaxial digits can result from ectopic polarizing activity due to mutations in regulatory regions that control expression of the *Shh* gene, specifically in the developing limb bud (Anderson et al 2012). Other changes in digit number may be the result of alterations in limb bud width and the length of the apical ectodermal ridge.

ANOMALIES OF THE DIGITS

Webbing of the fingers is the most common congenital anomaly of the hand. It occurs because of failure to remove cells between the digit rays during stages 19–23. Clinodactyly is a congenital condition in which the little finger is curved towards the ring finger. It can occur in isolation, or in association with chromosomal abnormalities like Down's syndrome. Camptodactyly is a congenital condition in which the little finger, and sometimes the ring finger, are held in a fixed, flexed position. Symphalangism is a rare congenital disorder in which there is fusion of the interphalangeal joints. The affected finger is stiff, skin creases are absent over the affected joint, there is minimal joint space, and sometimes the digit is shorter than normal. Several different types of brachydactyly, shortening of the digits, can be recognized according to which digits and which parts of these digits are affected. The genetic basis of different types of brachydactyly has now been determined, which is particularly gratifying because brachydactyly was the first condition in humans to be recognized as a dominant Mendelian condition (Gao and He 2004).

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